Local Projections or VARs: Evidence from Monte-Carlo Simulations

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Motivation

- VARs and Local Projection used for Impulse Response Analysis, think Unemployment and Interest Rate, Oil Prices and Inflation, ...
- finite sample comparison (VAR(p) vs. LP(p))
  1. In finite samples, bias-variance puzzle
  2. VARs more efficient, LPs more robust to misspecification
- In Population both estimate the same Impulse Response
- Ludwig identity shows that equal lag comparison is inherently flawed

(BL:) Where does a fixed LP(4) sit on the VAR complexity frontier and does the conventional equal-lag ranking sruvive?
(EX:) Does any genuine LP advantage emerge once misspecification is introduced?

Agenda

1. Theory
2. DGP
3. Results
4. Discussion of Results
5. Conclusion

VARs

$$Y_t = \nu + \mathbf{A} Y_{t-1} + U_t \xRightarrow[restr.K]{A1} y_t = \mu + \sum_{\ell=0}^{\infty} \Psi_\ell \, u_{t-\ell}, \qquad \Psi_\ell := [\mathbf{A}^\ell] \tag{1}$$

Identification: As $\sum_u = \mathbb{E}[u_t u_t'] \neq I_K$, Define $u_t = B\varepsilon_t$ where $\varepsilon_t \sim (0, I_k)$

$$y_{t+h} = \mu + \sum_{\ell=0}^{\infty} \Theta_\ell \, \varepsilon_{t+h-\ell} \tag{2}$$

Suffer from

- LS bias compunds as horizon grows Kilian 1998
- near-unit root behavior
- misspecification through MA dynamics or lag order

LPs

$$y_{t+h} = \alpha_h + \beta_h \, s_t + \boldsymbol{\gamma}_h' \, \mathbf{x}_t + \xi_{t+h}^{(h)}, \qquad h = 0, 1, \ldots, H, \tag{3}$$

Equivalence and Ludwig identity

DGPs

Baseline Results - Complexity Frontier

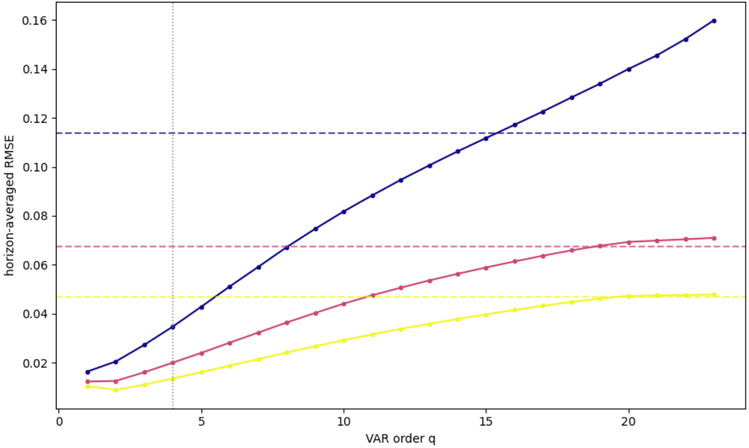

Figure 1: Complexity Frontier $\rho = 0.5$

Complexity Frontier - varying ρ

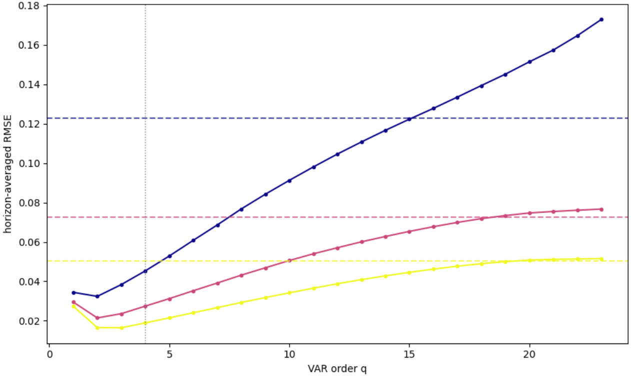

Figure 2: Complexity Frontier $\rho = 0.7$

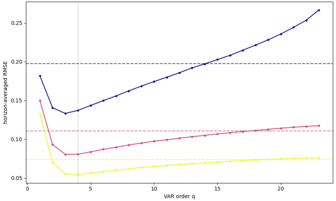

Figure 3: Complexity Frontier $\rho = 0.95$

Complexity Frontier - varying ρ

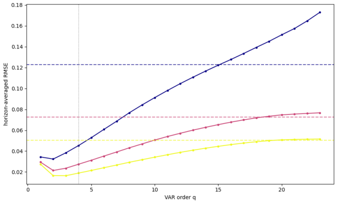

Figure 2: Complexity Frontier $\rho = 0.7$

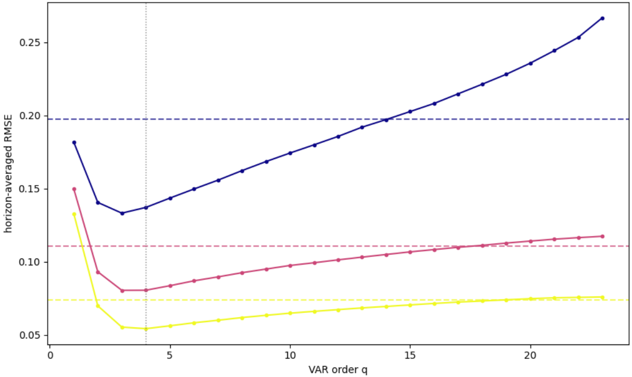

Figure 3: Complexity Frontier $\rho = 0.95$

RMSE and Coverage

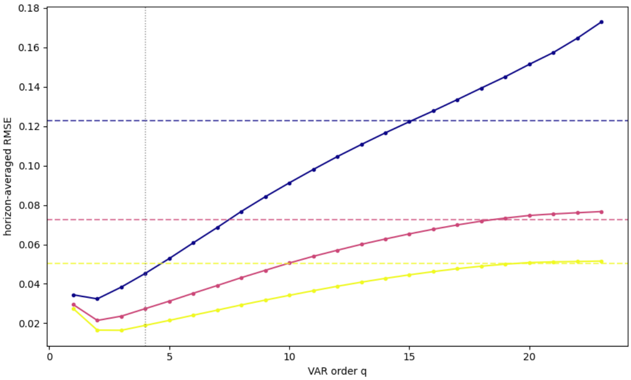

Figure 4: Kausaleffekte von Mentoring auf Sek. I Zuordnung

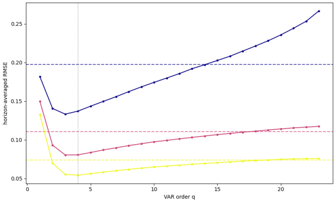

Figure 5: Vergleich Treatment mit Kontrollgruppen

Introducing Misspecification (1)

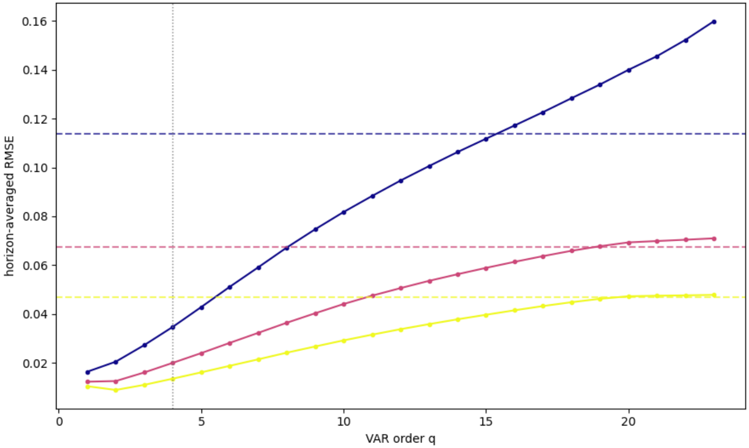

Figure 6:

Introducing Misspecification (2)

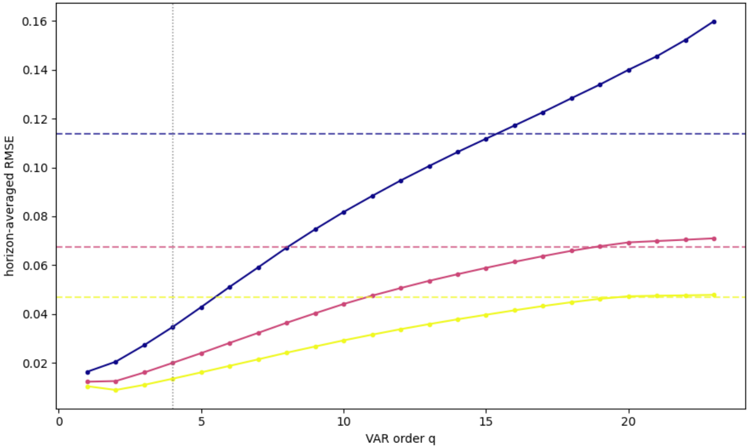

Figure 7:

Discussion of Results

- As T decreases, LP(4) is equal to a less parsimonous VAR $\rightarrow$ relatively better
- As ρ increases, LP(4) is equal to a less parsimonous VAR $\rightarrow$ relatively better
- As τ increases, LP(4) is equal to a less parsimonous VAR $\rightarrow$ relatively better
- At equal lags, VAR(p) is preferable choice if applicable, except for coverage
- higher order VARs closes coverage gap, at cost of bias and variance

Conclusion

1.
2.
3.

Vielen Dank für Ihre Aufmerksamkeit!